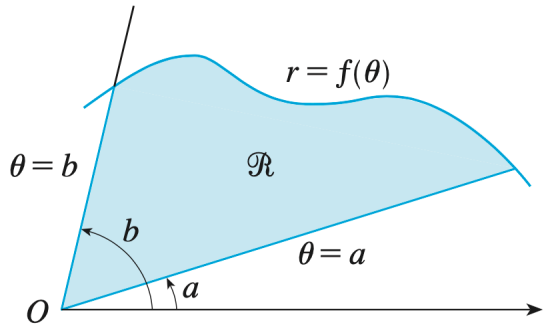
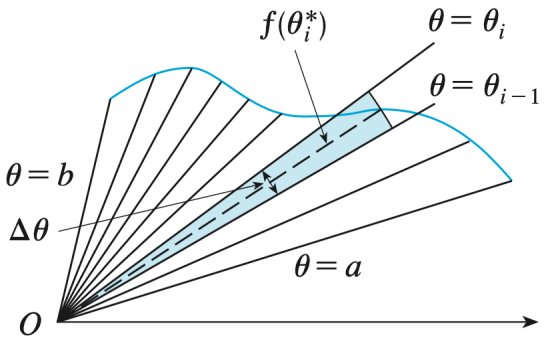
We will determine a formula for the area of a region $\mathscr{R}$ bounded by rays $\theta = a$, $\theta = b$, and by a polar curve $r = f(\theta)$, where f is a positive continuous function and where $0 < b - a \leq 2\pi$.



1. Divide the interval $[a,b]$ into subintervals with endpoints $\theta_0 < \theta_1 < \theta_2 < \cdots < \theta_n$ of equal width $\Delta\theta = \theta_i - \theta_{i-1}$.
2. If we choose $\theta_i^* \in [\theta_{i-1}, \theta_i]$, then the area ΔA_i of the ith region is approximately the area of the sector of a circle with central angle $\Delta\theta$ and radius $f(\theta_i*)$: $\Delta A_i \approx \frac{1}{2}(f(\theta_i*))^2\Delta\theta$.
3. An approximation for the total area is $A \approx A_n = \sum_{i=1}^{n} \Delta A_i = \sum_{i=1}^{n} \frac{1}{2}(f(\theta_i*))^2\Delta\theta$.
4. The approximation improves as $n \to \infty$, and we define the area as

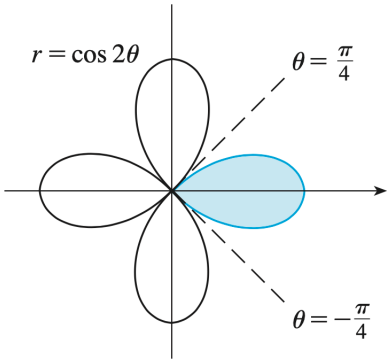
$$A = \lim_{n\to\infty} A_n = \lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{2}(f(\theta_i*))^2\Delta\theta = \int_a^b \frac{1}{2}(f(\theta))^2 d\theta.$$



Since $r = f(\theta)$, we write the area of the region $\mathscr{R}$ as

$$\boxed{A = \int_a^b \frac{1}{2} r^2 d\theta}$$

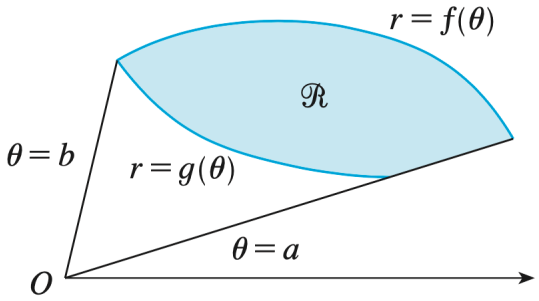
Example. Find the area enclosed by one loop of the four-leaved rose $r = \cos(2\theta)$.



Solution. By symmetry, we can choose any loop. As θ varies from $-\frac{\pi}{4}$ to $\frac{\pi}{4}$, r goes from 0 back to 0, so we sweep out one loop. By the formula above, this loop has area

$$\begin{aligned}
A &= \frac{1}{2}\int_{-\pi/4}^{\pi/4} \cos^2(2\theta)\,d\theta \\
&= \int_{0}^{\pi/4} \cos^2(2\theta)\,d\theta \\
&= \int_{0}^{\pi/4} \frac{1}{2}(1+\cos(4\theta))\,d\theta \\
&= \frac{1}{2}\int_{0}^{\pi/4} d\theta + \frac{1}{2}\int_{0}^{\pi/4} \cos(4\theta)\,d\theta \\
&= \frac{1}{2}\int_{0}^{\pi/4} d\theta + \frac{1}{8}\int_{0}^{\pi} \cos(u)\,du \\
&= \frac{\pi}{8} + \frac{1}{8}\sin(u)\Big|_0^{\pi} \\
&= \frac{\pi}{8} + \frac{1}{8}(\sin(\pi) - \sin(0)) \\
&= \frac{\pi}{8} + \frac{1}{8}(0) \\
&= \boxed{\frac{\pi}{8}}.
\end{aligned}$$

Now consider a region $\mathscr{R}$ bounded by rays $\theta = a$ and $\theta = b$, and two polar curves with polar equations $r = f(\theta)$, $r = g(\theta)$, where $f(\theta) \geq g(\theta) \geq 0$ and $0 < b - a \leq 2\pi$.



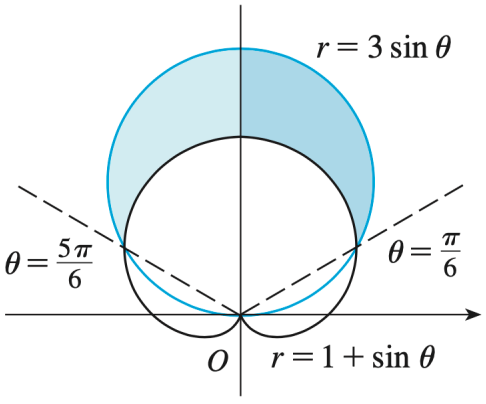
The area A of $\mathscr{R}$ is found by subtracting the area inside $r = g(\theta)$ from the area inside $r = f(\theta)$, so we have

$$A = \int_a^b \frac{1}{2}(f(\theta))^2 d\theta - \int_a^b \frac{1}{2}(g(\theta))^2 d\theta$$
$$= \boxed{\frac{1}{2}\int_a^b [f(\theta))^2 - g(\theta))^2] d\theta}.$$

Remark. The fact that a point has many representations in polar coordinates sometimes makes it difficult to find all the points of intersection of two polar curves. It is therefore recommended to plot the graphs of both curves in order to find all points of intersection.

Example. Find the area of the region that lies inside the circle $r = 3\sin(\theta)$ and outside the cardioid $r = 1 + \sin(\theta)$.

Solution.



The curves intersect when $3\sin(\theta) = 1 + \sin(\theta) \iff \sin(\theta) = \frac{1}{2} \iff \theta = \frac{\pi}{6}, \frac{5\pi}{6}$. Therefore

$$A = \tfrac{1}{2}\int_{\pi/6}^{5\pi/6} (3\sin(\theta))^2 d\theta - \tfrac{1}{2}\int_{\pi/6}^{5\pi/6} (1 + \sin(\theta))^2 d\theta.$$

Since the region is symmetric about the vertical axis $\theta = \frac{\pi}{2}$, we can write

$$\begin{aligned}
A &= 2\left[\tfrac{1}{2}\int_{\pi/6}^{\pi/2} 9\sin^2(\theta)d\theta - \tfrac{1}{2}\int_{\pi/6}^{\pi/2}(1+2\sin(\theta)+\sin^2(\theta))d\theta\right] \\
&= \int_{\pi/6}^{\pi/2}(8\sin^2(\theta)-1-2\sin(\theta))d\theta \\
&= \int_{\pi/6}^{\pi/2}(3-4\cos(2\theta)-2\sin(\theta))d\theta \qquad \text{because } \sin^2(\theta)=\tfrac{1}{2}(1-\cos(2\theta)) \\
&= 3\theta - 2\sin(2\theta)+2\cos(\theta)\Big|_{\pi/6}^{\pi/2} \\
&= \boxed{\pi}.
\end{aligned}$$

Arc Length. Recall that the arc length of a parametric curve between $a \le \theta \le b$ is given by

$$L = \int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta.$$

Using our formulas for $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ from before, this can be written as

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

Example. Find the length of the cardioid $r = 1+\sin(\theta)$.
Solution. The full length of the cardioid is given by the parameter interval $0 \le \theta \le 2\pi$. Using the formula above,

$$\begin{aligned}
L &= \int_0^{2\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \\
&= \int_0^{2\pi} \sqrt{(1+\sin(\theta))^2 + \cos^2(\theta)}\, d\theta \\
&= \int_0^{2\pi} \sqrt{2+\sin(\theta)}\, d\theta \\
&= 2\int_{-\pi/2}^{\pi/2} \sqrt{2+\sin(\theta)}\, d\theta \qquad \text{(by symmetry)} \\
&= 2\int_{-\pi/2}^{\pi/2} \sqrt{2+\sin(\theta)} \times \frac{\sqrt{2-2\sin(\theta)}}{\sqrt{2-2\sin(\theta)}}\, d\theta \\
&= 2\int_{-\pi/2}^{\pi/2} \frac{\sqrt{4-4\sin^2(\theta)}}{\sqrt{2-2\sin(\theta)}}\, d\theta \\
&= 4\int_{-\pi/2}^{\pi/2} \frac{\cos(\theta)}{\sqrt{2-2\sin(\theta)}}\, d\theta \qquad \text{(since } \cos(\theta) > 0 \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}\text{)}.
\end{aligned}$$

Let $u = 2 - 2\sin(\theta)$. Then $\frac{du}{-2} = \cos(\theta)$ and the integral above becomes

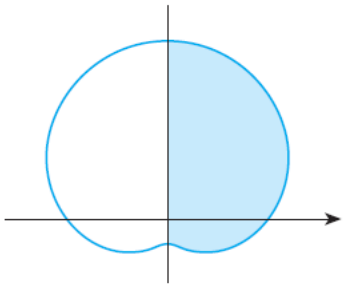
$$\begin{aligned} L &= \frac{4}{-2}\int_4^0 u^{-1/2}du \\ &= -2(2)u^{1/2}\Big|_4^0 \\ &= (-4)(-\sqrt{4}) \\ &= 4(2) \\ &= \boxed{8}. \end{aligned}$$

Exercises

1. Find the area of the region that is bounded by the given curve and lies in the specified sector:

$$r = e^{-\theta/4}, \quad \pi/2 \le \theta \le \pi.$$

2. Find the area of the shaded region



$r = 4 + 3\sin\theta$

3. Find the area of the region enclosed by one loop of $r = \sin(4\theta)$.

4. Find the area of the region that lies inside the first curve and outside the second curve: $r = 2\cos(\theta),\ r = 1$.

5. Find the area inside the larger loop and outside the smaller loop of the limacon $r = \frac{1}{2} + \cos(\theta)$.

6. Find the length of the polar curve $r = 2\cos(\theta),\ 0 \le \theta \le \pi$.

Answers. 1.) $e^{-\pi/4} - e^{-\pi/2}$, 2.) $\frac{41\pi}{4}$, 3.) $\frac{\pi}{3} + \frac{\sqrt{3}}{2}$, 4.) $\frac{\pi}{16}$, 5.) $\frac{1}{4}(\pi + 3\sqrt{3})$, 6.) 2π.